

CS 535: Assignment 02

Programming Assignments (90%)

- **This assignment is in C++.**
- Copy `src/app/Assign01.cpp` and name it **`src/app/Assign02.cpp`**
- Change the application name from "Assign01" to "Assign02"
- Change the window title to be "Assign02: " + your SITNET ID
 - o E.g., "Assign02: realemj"
- Make a copy of the `vulkanshaders/Assign01` folder and name it **`vulkanshaders/Assign02`**
- Modify **`CMakeLists.txt`** by adding the following line to the end of the file:
 - o `CREATE_VULKAN_EXECUTABLE(Assign02)`
- Make sure the program configures, compiles, and runs as-is
- Create a function of your own design:
`pro::HostMesh<ProVertex> createShape(int maxSub = 10, bool flipWinding = false)`
 - o This function should take at least two parameters:
 - **`int maxSub`**
 - This number controls how tessellated (i.e., how dense the triangles are in) your output shape is. At least, # of triangles $\geq$ `maxSub`.
 - **`bool flipWinding`**
 - If false (default), use counter-clockwise ordering. **If true, flip to clockwise (THIS SHOULD MAKE ALL YOUR TRIANGLES DISAPPEAR).**
 - You can add as many additional parameters as you like to control your shape.
 - o **Your function should NOT use globals.**
 - o This function should return a `pro::HostMesh<ProVertex>` mesh with the following constraints:
 - It should render properly (obviously).
 - It should be algorithmically/mathematically generated in some fashion.
 - It should use `maxSub` to control how many triangles are being generated.
 - All vertices MUST be STRICTLY INSIDE the window (so all x and y values should be between -1.0 and 1.0).
 - It should ensure that all triangles use a CONSISTENT winding order.
 - All vertices should have `z = 0.5f`.
 - At least some of the vertices should be different colors.
 - Alpha channel of colors should always be 1.0f.
 - Color values should be between 0.0f and 1.0f.
 - Triangles should NOT overlap.
 - Be creative 😊

- In the **main()** function, make the following changes:
 - o Comment out the line that adds simpleQuad to the vector allHostMeshes.
 - o Add your shape to allHostMeshes.
 - o **CORRECTION**: Ensure that this line in the loop that creates Vulkan meshes looks like this:
 - **allMeshes[i] = pro::createVulkanMesh(vkInitData, allHostMeshes[i], false);**
- You can choose to leave your background animation on or off.

Screenshots (5%)

- **Take TWO screenshots** of the application window.
 - o Use two different combinations of parameters.
 - o Two different values of maxSub is an acceptable variation.
 - o Do NOT set flipWinding to true for the screenshots.
- The naming convention of the screenshots should be "**Assign02_<params>.png**", where <params> is replaced with the parameters used.
 - o E.g., Assign02_maxSub_30.png
- Copy the images into the **screenshots** folder.

README (5%)

- Under the "Applications" section, add a subsection (same format as Assign01) very briefly describing what you decided to do for your shape. If you added additional parameters, briefly explain them here.

Grading

Your OVERALL assignment grade is weighted as follows:

- 90% - Programming
- 5% - Screenshots
- 5% - README

I reserve the right to take points off for not meeting the specifications in this assignment description.

The following specific penalties (applied to the WHOLE assignment) can be expected:

Problem	Maximum Penalty
Code that does not compile	60
Shape not rendering properly	60
Shape purely hardcoded and not using parameters properly	30
Winding order inconsistent	10
Vertices outside of or on very edge of window	10

Overlapping triangles	10
Bad/out-of-range background color values (including alpha)	10
Usage of globals in function instead of parameters	10
Wrong z value for coordinates	5
Vertices all the same color	5
Incorrect application name	5
Incorrect window title	5
Technically correct code, but poor code design	5
Missing assignment Git branch	5
Missing shaders folder	10
Missing screenshots	5
README update missing/incorrect	5

In general, these are things that will be penalized:

- **Code that does not compile (up to 60 points off!)**
- Sloppy or poor coding style
- Bad coding design principles
- Code that crashes, does not run, or takes a VERY long time to complete
- Using code from ANY source other than the course materials
- Collaboration on code of ANY kind; this is an INDIVIDUAL PROJECT
- Sharing code with other people in this class or using code from this or any other related class
- Output that is incorrect
- Algorithms/implementations that are incorrect
- Submitting improper files
- Failing to submit ALL required files